TAMID QUANT — STRATEGY ONE PAGER

Bitcoin Digital Options: Kalshi Price vs. Model Probability

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The Idea

Kalshi lists hourly "Bitcoin above $K at 5pm ET" contracts that pay $1 if true. Each is a digital option, so its fair price is the probability that BTC finishes above K. We estimate that probability from a volatility forecast, trade the contract when it is more than 4c away from the model, and optionally hedge the BTC delta. Tested on 3,387 contracts over 120 expiries (Jun to Oct 2026): the market is already well priced, and no edge survives costs (see Key Risks).

Mathematical Framework

Core Model: *Digital pricing. With S the spot price, σ the forecast volatility and h the hours to expiry, $p = P(S_T > K) = N(d2)$, $d2 = [\ln(S/K) - \sigma^2 h/2] / (\sigma\sqrt{h})$. The hedge ratio is the delta $\Delta = \varphi(d2) / (S \cdot \sigma\sqrt{h})$.*

Signal Construction: *Edge = model p − Kalshi price, using only data up to the signal time:*

> σ^2 from an EWMA of hourly returns ($\alpha = 0.03$); GARCH(1,1) and 168-hour realized vol as alternatives. Tails from filtered historical simulation: the empirical distribution of the past 90 days of standardized returns, kernel-smoothed, replacing the normal. Quality check: Brier score of model vs. market price.

Entry / Exit Logic: *Signal 6 hours before expiry, fill one hour later at the Kalshi bid or ask:*

> If p − ask > 4c, buy YES; if bid − p > 4c, sell YES. Hold to settlement. The hedged variant also holds −(side)·Δ in BTC, rebalanced hourly at a 5bps cost on turnover. Strikes within ±6% of spot, at most 30 contracts per expiry.

Position Sizing: *One contract per qualifying strike. Kalshi collateral is at most $1 per contract, and the hedge holds about $51 of BTC per $1 contract.*

Key Assumptions

- Short-horizon BTC volatility and tails resemble the past 90 days
- Hourly-candle bid/ask quotes are fillable, with fee $0.07 \cdot P \cdot (1 - P)$ (assumed; no depth data)
- Each expiry day is one independent draw, so the effective sample is about 120 days, not 3,387 contracts

Key Risks

- No pricing edge: Brier score is 0.0416 for the market and 0.0416 for the model (difference +0.5e-4, 95% CI −9 to +10 e-4). Two of three normal-tail variants are significantly worse
- Unhedged P&L is +2.2c per contract over 386 trades (95% CI −1.1c to +5.6c), but in-sample is −0.8c, the 3h and 12h horizons are negative, and GARCH is negative: one positive cell out of many tried
- The hedge cuts the standard deviation from 29c to 23c but costs 5c per contract at 5bps (−5.6c net, still negative at 2bps). The only robust pattern: 10-20c contracts are overpriced (5.7% won)

At a Glance

Asset Class:	*Crypto / Prediction Markets*
Strategy Type:	*Model vs. Market Mispricing*
Direction:	*Long/Short Digitals (+ Optional BTC Hedge)*
Holding Period:	*Hours (Hold to Expiry)*
Universe:	*Kalshi KXBTCD vs. Coinbase BTC-USD*

Data Requirements

- Kalshi KXBTCD hourly bid, ask, volume and settlement — public API
- Coinbase BTC-USD hourly candles for volatility and the hedge — public API
- 3,387 settled contracts, 120 expiries (5pm ET), 51,149 hourly quotes
- Kalshi fee schedule; order-book depth (not yet collected)

Key Parameters (starting values)

- Edge threshold: 4c (tested 2c to 10c)
- Signal horizon: 6h before expiry (tested 3h, 12h)
- Volatility: EWMA $\alpha = 0.03$; FHS 90-day window
- Hedge cost: 5bps (tested 0 to 10bps); fee $0.07 \cdot P \cdot (1 - P)$

Open Questions

- Do the intraday contracts mis-price after sharp BTC moves, when the quote lags spot? Needs minute-level data
- Can resting (maker) orders avoid the spread and fee that erased the edge?
- Is the overpricing of 5-20c contracts a stable longshot bias, and is selling them worth the tail risk?
- Is there a Kalshi vs. Polymarket gap for BTC?